

Snippets from across the world

While this article has focused on the Indian landscape, it would be incomplete without a mention of recent international innovations in this space. Take a look:

iRobot's medical robots. Recently approved by the FDA, iRobot series of RP-VITA robots lets doctors interact with hospital patients remotely. It can navigate around hospitals on its own and has a 38cm (15-inch) LCD screen as the face. There is an associated app running on the doctor's smartphone, through which the doctor and patients can interact any time.

SCOUT. Developed by NASA-based start-up Scanadu, SCOUT studies the vital signs that reflect your health condition in less than ten seconds. The information is transmitted to a mobile app that instantly calculates the pulse, heart rate, electrical heart activity, temperature, heart rate variability and blood oxygenation. This information can be shared with a doctor, or stored and studied over time to figure out how healthy you are. The device uses body sensors, molecular diagnostics, imaging and sound analysis, data analytics and a suite of algorithms to study and offer a comprehensive, real-time picture of your health data. They fit the description of a 'tricorder.' The software platform is extensible and can be opened up to input from other tracking and monitoring devices.

iCouch. This is an online platform that connects people to specialist doctors in the comfort and safety of their own homes. iCouch has a panel of verified therapists. Using your laptop or mobile, you can access icouch.com, find a therapist, create an account and buy session credits, check the therapist's schedule and fix the time, log on at the confirmed time and video-chat with the specialist. The sessions are confidential and protected using several layers of security.

Magic Carpet by GE and Intel. Last year, GE and Intel collaborated on a sensor-fitted home carpet that helps watch the movement of elders or children and inform caretakers when they fall. The carpet consists of special plastic optical fibres and sensors that map 2D images of a patient's footprints by using light propagating under the surface, similar to tomographic techniques used by medical scanners. These images can be analysed to detect patterns and changes in a patient's walking habits and predict whether he is at an increased risk of falling. It can also detect gait changes, weight changes, etc.

Examiner. An iPhone adaptor recently approved by the FDA helps doctors to remotely examine a patient's eye, using an ophthalmoscope attached to an iPhone. There is an associated application as well.

equipment, communicating health status to doctors and receiving advice from them has expanded the scope of telemedicine and health monitoring.

From being used only for delivering medical services remotely to rural areas initially, these technologies are now also being used to provide better services to patients in hospitals, take care of specially-challenged or aged people, monitor foetal health and even improve the fitness level of people.

Let us take a look at the key components of telemedicine, as well as recent innovations and some promising technologies and products in this space.

What's inside?

There is a wide array of telemedicine and health monitoring equipment available today. Broadly, we can divide them into diagnostic and therapeutic equipment like electrocardiographs and pacemakers, imaging technologies like X-ray and magnetic resonance imaging (MRI), medical instruments like blood analysis systems and dialysis systems, patient monitoring and consumer products like remote monitoring tools, insulin pumps and heart-rate monitors, as well as wellness equipment like pedometers and cardio trainers.

Such equipment can be thought of as having a layer of information

and communication technology (ICT) atop a medical device. These usually comprise a combination of sensors, MEMS, low-power high-performance microcontrollers, wireless modules, signal conditioners, analogue-to-digital converters, etc, which work alongside user-friendly but powerful software. The infrastructure required includes uninterrupted power supply and Internet—both of which are a challenge in the Indian landscape.

"With increasing penetration of mobile infrastructure into the remote areas, mobility is expected to play a larger role in healthcare as well. Solutions will have to be designed to be mobile, as we go along. 3.5G and 4G are going to have a huge impact on the ability to deliver healthcare at the patients' doorsteps. However, until the connectivity reaches rural areas, the impact will be seen only on the urban population and the changes will miss the vast majority of the population that lives in the rural areas," remarks Rajeev Kumar, founder director, Neuro-synaptic Communications—a company that has made a mark in the Indian telemedicine arena with its ReMeDi solution.

"Availability of electricity is also a huge challenge in rural as well as semi-urban areas. Alternatives like solar are still too expensive, although the prices are coming down. Plus, operation,

maintenance and support of the telemedicine infrastructure are always a problem in the absence of adequately trained personnel," he adds.

These challenges reflect in the design and engineering goals of telemedicine and health monitoring equipment.

Design considerations and constraints

According to Kumar, when developing telemedicine solutions, engineers should keep in mind data accuracy, patient safety, low power consumption, ease of use and ruggedness. At the same time, these solutions have to be very powerful and accurate.

Software engineers also have to consider the rural environment when developing applications for remote healthcare. The software has to be light, capable of running on minimal resources. Considering the fact that it might be used by untrained professionals, the interface must be very simple. It must also support local languages.

Gupta adds, "While implementing ICT-based solutions, especially in rural areas of India, the designers will have to keep in mind that the infrastructure available is at the bare minimum. The solution should be capable of adapting to this infrastructural deficit and work its way around enabling the healthcare ser-